

★ C.2-6

Show how to construct a set of n events that are pairwise independent but such that no subset of $k > 2$ of them is mutually independent.

★ C.2-7

Two events A and B are *conditionally independent*, given C , if

$$\Pr \{A \cap B \mid C\} = \Pr \{A \mid C\} \cdot \Pr \{B \mid C\}.$$

Give a simple but nontrivial example of two events that are not independent but are conditionally independent given a third event.

★ C.2-8

Professor Gore teaches a music class on rhythm in which three students—Jeff, Tim, and Carmine—are in danger of failing. Professor Gore tells the three that one of them will pass the course and the other two will fail. Carmine asks Professor Gore privately which of Jeff and Tim will fail, arguing that since he already knows at least one of them will fail, the professor won't be revealing any information about Carmine's outcome. In a breach of privacy law, Professor Gore tells Carmine that Jeff will fail. Carmine feels somewhat relieved now, figuring that either he or Tim will pass, so that his probability of passing is now $1/2$. Is Carmine correct, or is his chance of passing still $1/3$? Explain.

C.3 Discrete random variables

A *(discrete) random variable* X is a function from a finite or countably infinite sample space S to the real numbers. It associates a real number with each possible outcome of an experiment, which allows us to work with the probability distribution induced on the resulting set of numbers. Random variables can also be defined for uncountably infinite sample spaces, but they raise technical issues that are unnecessary to address for our purposes. Therefore we'll assume that random variables are discrete.

For a random variable X and a real number x , we define the event $X = x$ to be $\{s \in S : X(s) = x\}$, and thus

$$\Pr\{X = x\} = \sum_{s \in S: X(s)=x} \Pr\{s\} .$$

The function

$$f(x) = \Pr\{X = x\}$$

is the **probability density function** of the random variable X . From the probability axioms, $\Pr\{X = x\} \geq 0$ and $\sum_x \Pr\{X = x\} = 1$.

As an example, consider the experiment of rolling a pair of ordinary, 6-sided dice. There are 36 possible outcomes in the sample space. Assume that the probability distribution is uniform, so that each outcome $s \in S$ is equally likely: $\Pr\{s\} = 1/36$. Define the random variable X to be the *maximum* of the two values showing on the dice. We have $\Pr\{X = 3\} = 5/36$, since X assigns a value of 3 to 5 of the 36 possible outcomes, namely, (1, 3), (2, 3), (3, 3), (3, 2), and (3, 1).

We can define several random variables on the same sample space. If X and Y are random variables, the function

$$f(x, y) = \Pr\{X = x \text{ and } Y = y\}$$

is the **joint probability density function** of X and Y . For a fixed value y ,

$$\Pr\{Y = y\} = \sum_x \Pr\{X = x \text{ and } Y = y\} ,$$

and similarly, for a fixed value x ,

$$\Pr\{X = x\} = \sum_y \Pr\{X = x \text{ and } Y = y\} .$$

Using the definition (C.16) of conditional probability on page 1187, we have

$$\Pr\{X = x \mid Y = y\} = \frac{\Pr\{X = x \text{ and } Y = y\}}{\Pr\{Y = y\}} .$$

We define two random variables X and Y to be **independent** if for all x and y , the events $X = x$ and $Y = y$ are independent or, equivalently, if for all x and y , we have $\Pr\{X = x \text{ and } Y = y\} = \Pr\{X = x\} \Pr\{Y = y\}$.

Given a set of random variables defined over the same sample space, we can define new random variables as sums, products, or other functions of the original variables.

Expected value of a random variable

The simplest, and often the most useful, summary of the distribution of a random variable is the “average” of the values it takes on. The *expected value* (or, synonymously, *expectation* or *mean*) of a discrete random variable X is

$$E[X] = \sum_x x \cdot \Pr\{X = x\} , \quad (\text{C.23})$$

which is well defined if the sum is finite or converges absolutely. Sometimes the expectation of X is denoted by μ_X or, when the random variable is apparent from context, simply by μ .

Consider a game in which you flip two fair coins. You earn \$3 for each head but lose \$2 for each tail. The expected value of the random variable X representing your earnings is

$$\begin{aligned} E[X] &= 6 \cdot \Pr\{2 \text{ H's}\} + 1 \cdot \Pr\{1 \text{ H}, 1 \text{ T}\} - 4 \cdot \Pr\{2 \text{ T's}\} \\ &= 6 \cdot (1/4) + 1 \cdot (1/2) - 4 \cdot (1/4) \\ &= 1. \end{aligned}$$

Linearity of expectation says that the expectation of the sum of two random variables is the sum of their expectations, that is,

$$E[X + Y] = E[X] + E[Y] , \quad (\text{C.24})$$

whenever $E[X]$ and $E[Y]$ are defined. Linearity of expectation applies to a broad range of situations, holding even when X and Y are not independent. It also extends to finite and absolutely convergent summations of expectations. Linearity of expectation is the key property that enables us to perform probabilistic analyses by using indicator random variables (see Section 5.2).

If X is any random variable, any function $g(x)$ defines a new random variable $g(X)$. If the expectation of $g(X)$ is defined, then

$$\mathbb{E}[g(X)] = \sum_x g(x) \cdot \Pr\{X = x\} .$$

Letting $g(x) = ax$, we have for any constant a ,

$$\mathbb{E}[aX] = a\mathbb{E}[X] . \quad (\text{C.25})$$

Consequently, expectations are linear: for any two random variables X and Y and any constant a ,

$$\mathbb{E}[aX + Y] = a\mathbb{E}[X] + \mathbb{E}[Y] . \quad (\text{C.26})$$

When two random variables X and Y are independent and each has a defined expectation,

$$\begin{aligned} \mathbb{E}[XY] &= \sum_x \sum_y xy \cdot \Pr\{X = x \text{ and } Y = y\} \\ &= \sum_x \sum_y xy \cdot \Pr\{X = x\} \Pr\{Y = y\} \quad (\text{by independence of } X \text{ and } Y) \\ &= \left(\sum_x x \cdot \Pr\{X = x\} \right) \left(\sum_y y \cdot \Pr\{Y = y\} \right) \\ &= \mathbb{E}[X] \mathbb{E}[Y] \quad (\text{by equation (C.23)}) . \end{aligned}$$

In general, when n random variables $X_1, X_2, \dots, X_n$ are mutually independent,

$$\mathbb{E}[X_1 X_2 \cdots X_n] = \mathbb{E}[X_1] \mathbb{E}[X_2] \cdots \mathbb{E}[X_n] . \quad (\text{C.27})$$

When a random variable X takes on values from the set of natural numbers $\mathbb{N} = \{0, 1, 2, \dots\}$, we have a nice formula for its expectation:

$$\begin{aligned} \mathbb{E}[X] &= \sum_{i=0}^{\infty} i \cdot \Pr\{X = i\} \\ &= \sum_{i=0}^{\infty} i \cdot (\Pr\{X \geq i\} - \Pr\{X \geq i+1\}) \\ &= \sum_{i=1}^{\infty} \Pr\{X \geq i\} , \end{aligned} \quad (\text{C.28})$$

since each term $\Pr\{X \geq i\}$ is added in i times and subtracted out $i - 1$ times (except $\Pr\{X \geq 0\}$, which is added in 0 times and not subtracted out at all).

A function $f(x)$ is *convex* if

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) \quad (\text{C.29})$$

for all x and y and for all $0 \leq \lambda \leq 1$. *Jensen's inequality* says that when a convex function $f(x)$ is applied to a random variable X ,

$$E[f(X)] \geq f(E[X]), \quad (\text{C.30})$$

provided that the expectations exist and are finite.

Variance and standard deviation

The expected value of a random variable does not express how “spread out” the variable’s values are. For example, consider random variables X and Y for which $\Pr\{X = 1/4\} = \Pr\{X = 3/4\} = 1/2$ and $\Pr\{Y = 0\} = \Pr\{Y = 1\} = 1/2$. Then both $E[X]$ and $E[Y]$ are $1/2$, yet the actual values taken on by Y are further from the mean than the actual values taken on by X .

The notion of variance mathematically expresses how far from the mean a random variable’s values are likely to be. The *variance* of a random variable X with mean $E[X]$ is

$$\begin{aligned} \text{Var}[X] &= E[(X - E[X])^2] \\ &= E[X^2 - 2XE[X] + E^2[X]] \\ &= E[X^2] - 2E[XE[X]] + E^2[X] \\ &= E[X^2] - 2E^2[X] + E^2[X] \\ &= E[X^2] - E^2[X]. \end{aligned} \quad (\text{C.31})$$

To justify the equation $E[E^2[X]] = E^2[X]$, note that because $E[X]$ is a real number and not a random variable, so is $E^2[X]$. The equation $E[XE[X]] = E^2[X]$ follows from equation (C.25), with $a = E[X]$. Rewriting equation (C.31) yields an expression for the expectation of the square of a random variable:

$$E[X^2] = \text{Var}[X] + E^2[X]. \quad (\text{C.32})$$

The variance of a random variable X and the variance of aX are related (see Exercise C.3-10):

$$\text{Var}[aX] = a^2 \text{Var}[X].$$

When X and Y are independent random variables,

$$\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y].$$

In general, if n random variables $X_1, X_2, \dots, X_n$ are pairwise independent, then

$$\text{Var} \left[\sum_{i=1}^n X_i \right] = \sum_{i=1}^n \text{Var} [X_i] . \quad (\text{C.33})$$

The *standard deviation* of a random variable X is the nonnegative square root of the variance of X . The standard deviation of a random variable X is sometimes denoted σ_X or simply σ when the random variable X is understood from context. With this notation, the variance of X is denoted σ^2 .

Exercises

C.3-1

You roll two ordinary, 6-sided dice. What is the expectation of the sum of the two values showing? What is the expectation of the maximum of the two values showing?

C.3-2

An array $A[1 : n]$ contains n distinct numbers that are randomly ordered, with each permutation of the n numbers being equally likely. What is the expectation of the index of the maximum element in the array? What is the expectation of the index of the minimum element in the array?

C.3-3

A carnival game consists of three dice in a cage. A player can bet a dollar on any of the numbers 1 through 6. The cage is shaken, and the payoff is as follows. If the player's number doesn't appear on any of the dice, the player loses the dollar. Otherwise, if the player's number appears on exactly k of the three dice, for $k = 1, 2, 3$, the player keeps the dollar and wins k more dollars. What is the expected gain from playing the carnival game once?

C.3-4

Argue that if X and Y are nonnegative random variables, then $E[\max \{X, Y\}] \leq E[X] + E[Y]$.

★ C.3-5

Let X and Y be independent random variables. Prove that $f(X)$ and $g(Y)$ are independent for any choice of functions f and g .

★ C.3-6

Let X be a nonnegative random variable, and suppose that $E[X]$ is well defined. Prove *Markov's inequality*:

$$\Pr \{X \geq t\} \leq E[X] / t \quad (\text{C.34})$$

for all $t > 0$.

★ C.3-7

Let S be a sample space, and let X and X' be random variables such that $X(s) \geq X'(s)$ for all $s \in S$. Prove that for any real constant t ,

$$\Pr \{X \geq t\} \geq \Pr \{X' \geq t\}.$$

C.3-8

Which is larger: the expectation of the square of a random variable, or the square of its expectation?

C.3-9

Show that for any random variable X that takes on only the values 0 and 1, we have $\text{Var}[X] = E[X] E[1 - X]$.

C.3-10

Prove that $\text{Var}[aX] = a^2 \text{Var}[X]$ from the definition (C.31) of variance.

C.4 The geometric and binomial distributions